

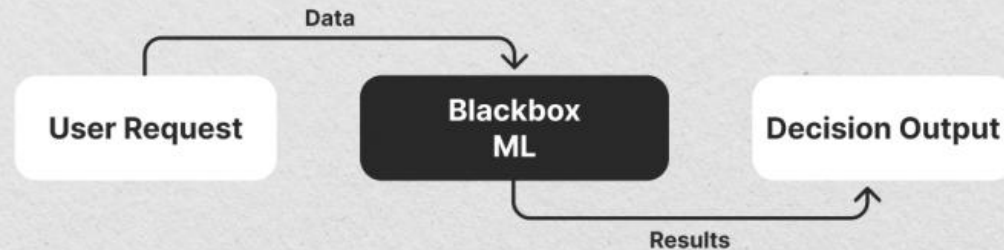
Explainable Artificial Intelligence (XAI)

DR. SIFAT MOMEN

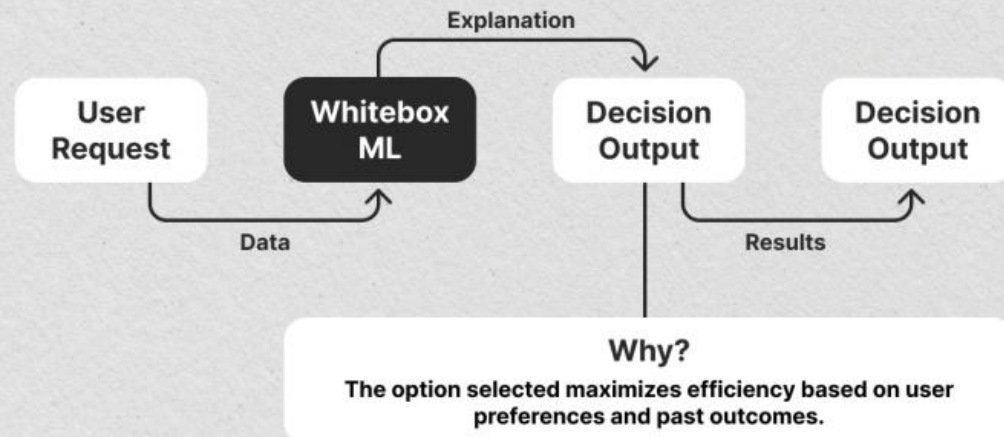
Blackbox versus Whitebox Machine Learning

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BLACKBOX MACHINE LEARNING



WHITEBOX MACHINE LEARNING



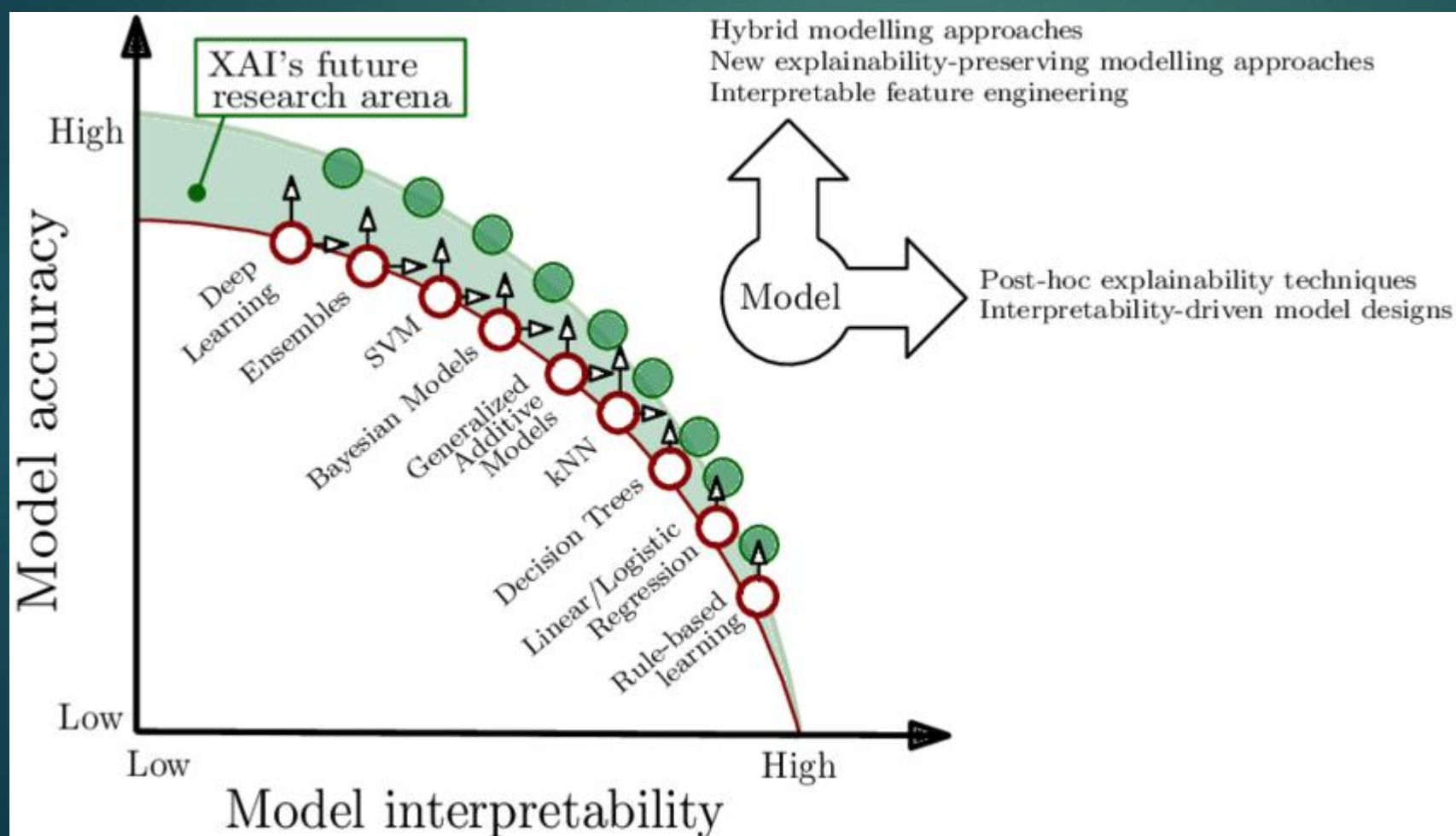
Explainability in AI

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- ▶ <<To describe the purpose, rationale, and decision-making process of the AI tool in a way that can be understood by the average person>>
 - ▶ Data Scientist
 - ▶ Domain Expert (medical doctor for instance)
 - ▶ User/customer
 - ▶ Regulator/lawyer/ ...

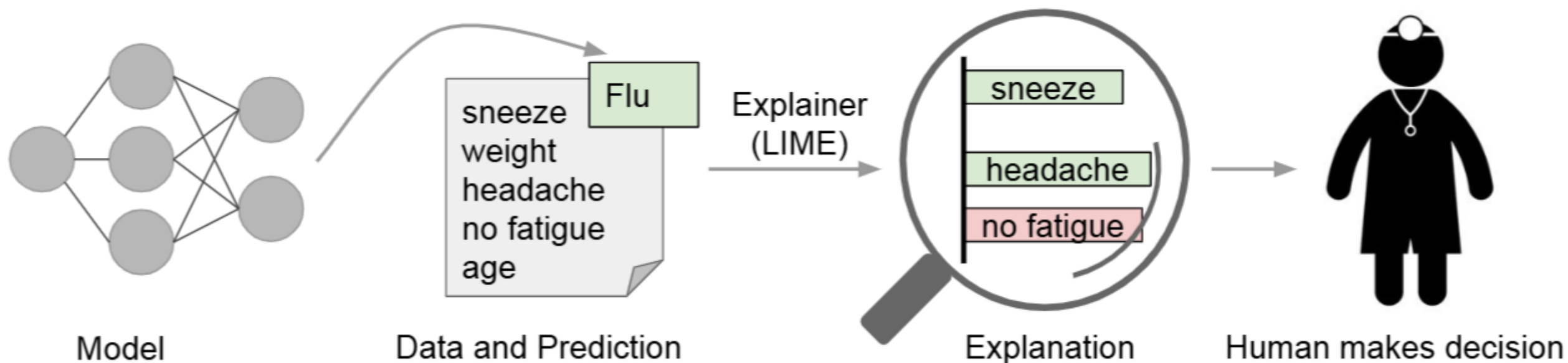
Performance and interpretability tradeoff

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Local Interpretable Model-Agnostic Explanations (LIME)

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Instance Specific Explanations

“Why Should I Trust You?”
Explaining the Predictions of Any Classifier

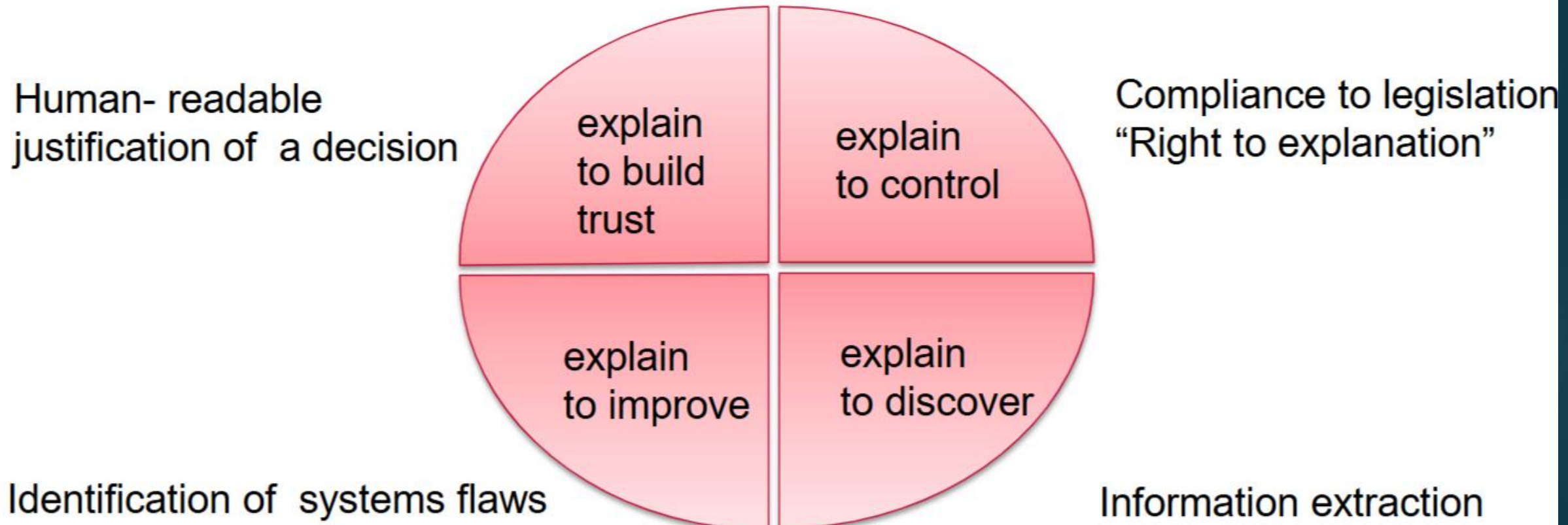
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The need for Explainability

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The need for Explainability

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RESEARCH-ARTICLE

Leakage in data mining: Formulation, detection, and avoidance

Authors:  [Shachar Kaufman](#),  [Saharon Rosset](#),  [Claudia Perlich](#),  [Ori Stitelman](#) | [Authors Info & Claims](#)

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Check for updates

A challenging example cited by Kaufman et al. is one where the patient ID was found to be heavily correlated with the target class in the training and validation data. This issue would be incredibly challenging to identify just by observing the predictions and the raw data, but much easier if explanations could have been provided.

The need for Explainability

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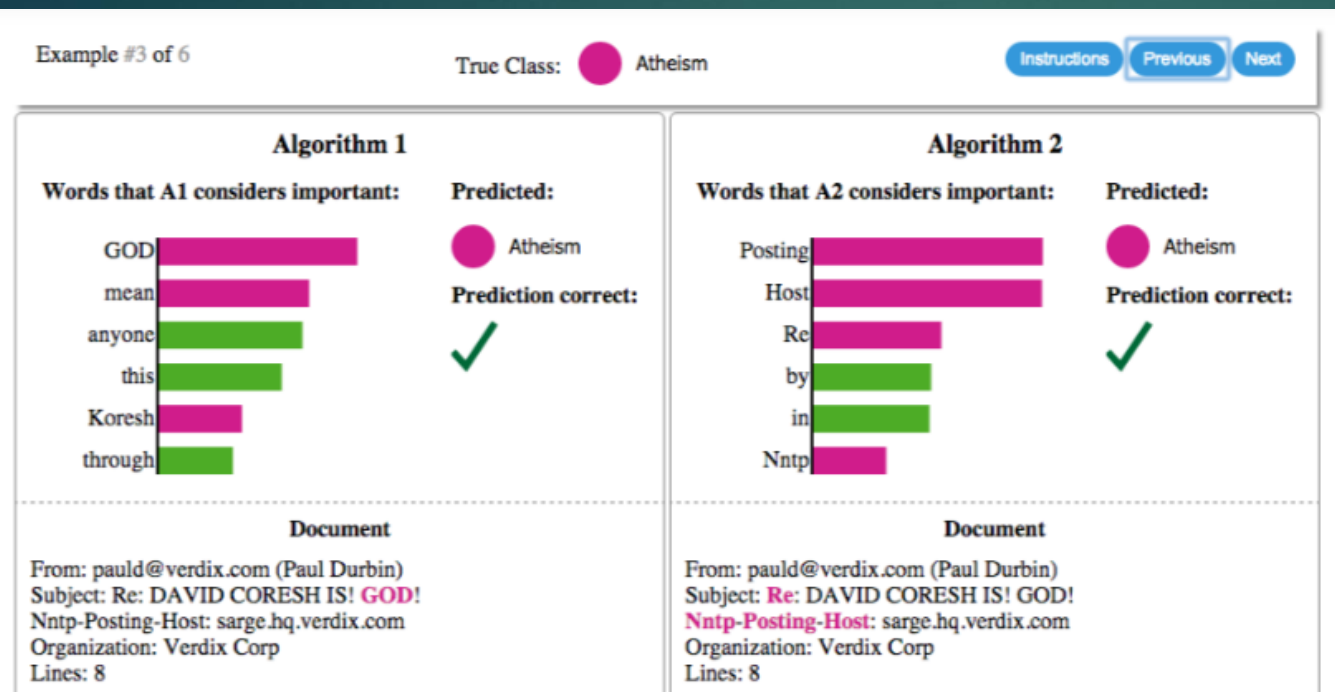


Figure 2: Explaining individual predictions of competing classifiers trying to determine if a document is about “Christianity” or “Atheism”. The bar chart represents the importance given to the most relevant words, also highlighted in the text. Color indicates which class the word contributes to (green for “Christianity”, magenta for “Atheism”).

Explainability can be very useful for model selection as well. The algorithm with higher accuracy on the validation set is actually much worse, a fact that is easy to see when explanations are provided (again, due to human prior knowledge), but hard otherwise.

Desired Characteristics for Explainers

- ▶ An essential criterion for explanations is that they must be **interpretable**, i.e., provide qualitative understanding between the input variables and the response.
- ▶ For example, if hundreds or thousands of features significantly contribute to a prediction, it is not reasonable to expect any user to comprehend why the prediction was made, even if individual weights can be inspected.
- ▶ This requirement further implies that explanations should be easy to understand.
- ▶ Finally, note that the notion of interpretability also depends on the target audience.

Desired Characteristics for Explainers

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- ▶ Another essential criterion is **local fidelity**.
- ▶ Although it is often impossible for an explanation to be completely faithful unless it is the complete description of the model itself, for an explanation to be meaningful it must at least be locally faithful, i.e. it must correspond to how the model behaves in the vicinity of the instance being predicted.
- ▶ Please note that local fidelity does not imply global fidelity: features that are globally important may not be important in the local context, and vice versa.

Desired Characteristics for Explainers

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- ▶ While there are models that are inherently interpretable, an explainer should be able to explain any model, and thus be **model-agnostic** (i.e. treat the original model as a black box).

How LIME Works

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1. **Pick an instance to explain**

You choose one particular prediction (e.g., why did the model predict "spam" for this specific email?).

2. **Generate perturbed samples around the instance**

LIME creates new samples by slightly tweaking the features of the original instance. For a text input, it might remove some words; for tabular data, it might tweak numerical values.

3. **Get predictions from the black-box model**

These perturbed samples are fed into the original model to see how the model behaves in the neighborhood of the instance.

4. **Weight the samples by similarity**

LIME assigns higher weights to perturbed samples that are more similar to the original instance (typically using Euclidean distance or cosine similarity).

5. **Train an interpretable model on the weighted samples**

LIME fits a simple, interpretable model (like a linear regression or decision tree) on the perturbed data using the weights. This model **approximates** the behavior of the black-box model **locally**.

6. **Present explanation**

The coefficients or rules from the simple model are then used as an explanation for the original prediction.

Example 1

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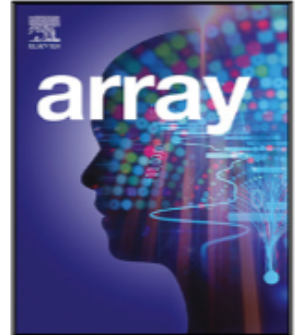


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AIDA: Artificial intelligence based depression assessment applied to Bangladeshi students

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Example 1

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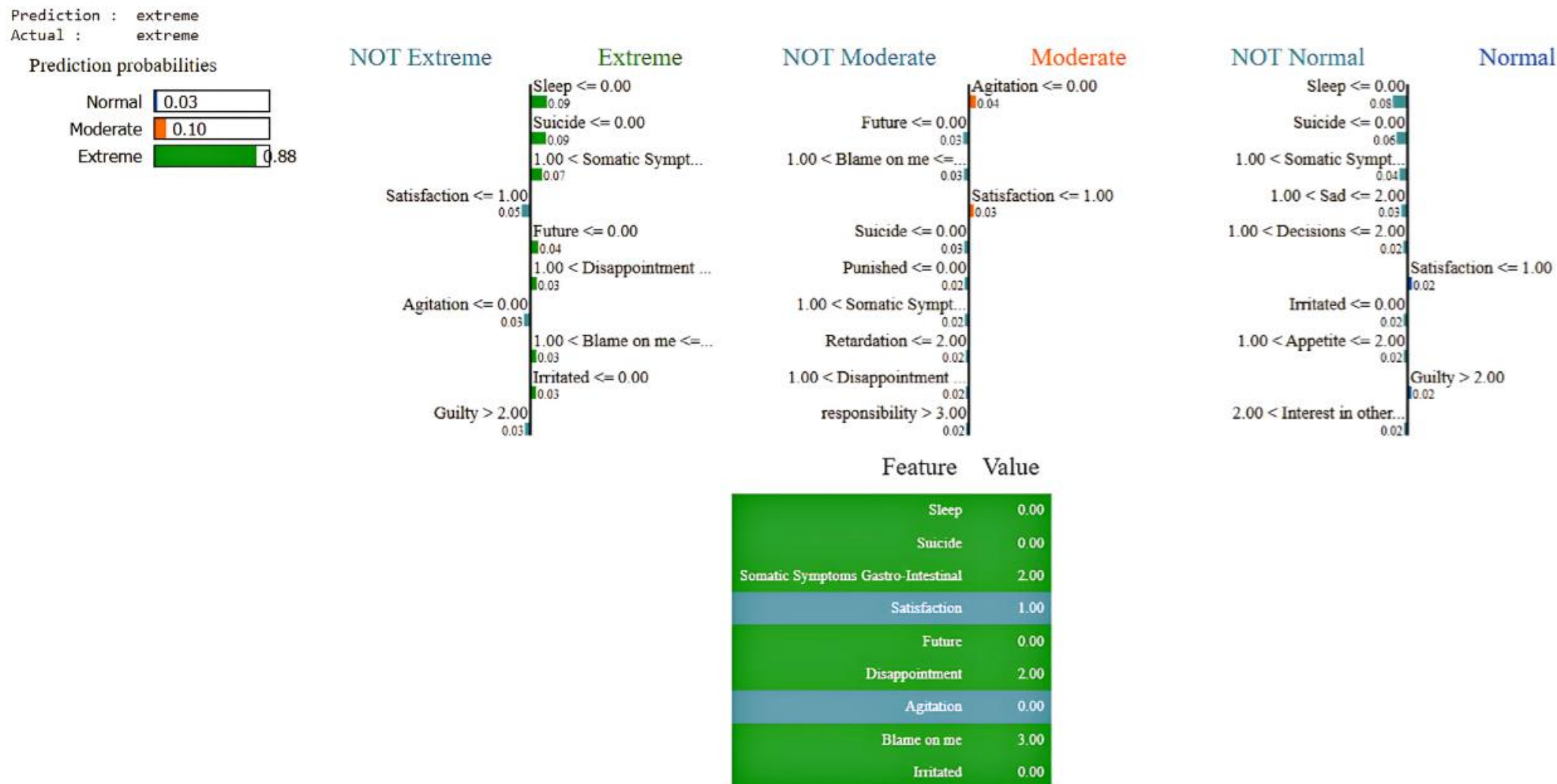


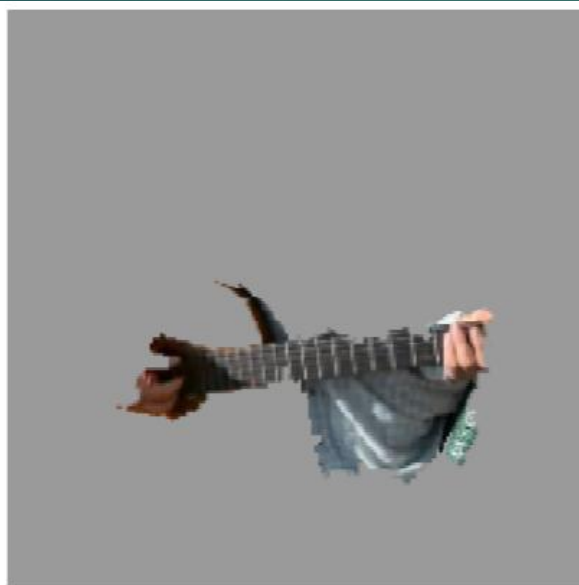
Fig. 13. Depression prediction interpretation of an extreme case of the LIME explainable AI.

Example 2

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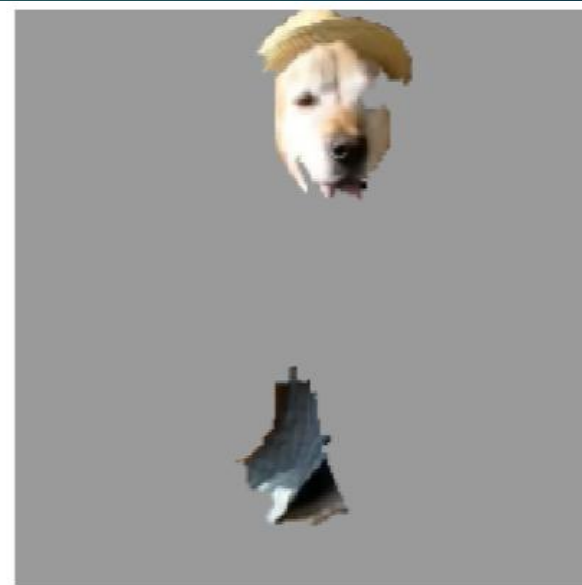
(a) Original Image



(b) Explaining *Electric guitar*



(c) Explaining *Acoustic guitar*



(d) Explaining *Labrador*

Figure 4: Explaining an image classification prediction made by Google's Inception neural network. The top 3 classes predicted are "Electric Guitar" ($p = 0.32$), "Acoustic guitar" ($p = 0.24$) and "Labrador" ($p = 0.21$)

LIME Code

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```
#!/pip install lime
```

```
import lime  
import lime.lime_tabular
```

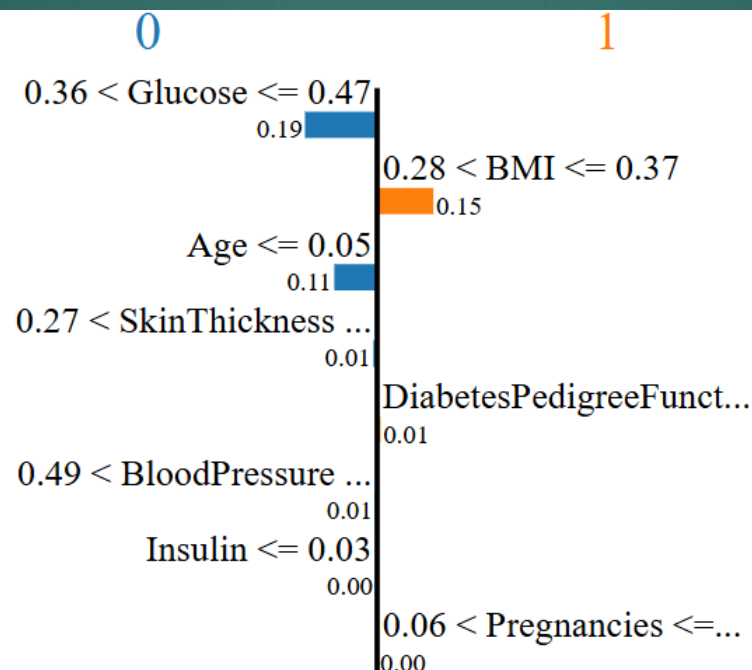
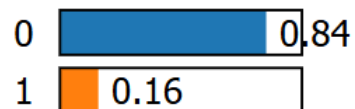
```
explainer = lime.lime_tabular.LimeTabularExplainer(training_data = X_train.values,  
mode='classification',  
feature_names=X_train.columns)
```

```
exp = explainer.explain_instance(X_test.iloc[1], dt.predict_proba)  
exp.show_in_notebook(show_table = True)
```

LIME Code

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Prediction probabilities



Feature	Value
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Glucose	0.44
BMI	0.36
Age	0.00
SkinThickness	0.44
DiabetesPedigreeFunction	0.15
BloodPressure	0.52
Insulin	0.03
Pregnancies	0.12